

Solution Requirements

DATE	02 JULY 2026
TEAM ID	
PROJECT NAME	A Comprehensive Measure of Well-Being: Human Development Index (HDI) Predictor
MAXIMUM MARKS	4 MARKS

1. Overview

This document specifies the functional and non-functional requirements for the Human Development Index (HDI) Prediction System. These requirements ensure that the system meets user needs, performs reliably, and is easy to maintain.

2. Functional Requirements (FR)

FR-1: Multi-Indicator Input Interface

- **Description:** The system must provide an input form that allows users to enter the five key development indicators: country category, life expectancy, expected years of schooling, mean years of schooling, and GNI per capita.
- **Priority:** High.
- **Verification:** User form loads successfully on the `/predict` route.

FR-2: Machine Learning Prediction

- **Description:** The system must load a pre-trained Linear Regression model to predict a continuous HDI score based on the user's inputs.
- **Priority:** High.
- **Verification:** Predicted scores are generated and displayed on the results card.

FR-3: Automatic Tier Classification

- **Description:** The system must automatically classify the predicted HDI score into one of the four standard UNDP development tiers:
- **Very High:** ≥ 0.800
- **High:** 0.700 to 0.799
- **Medium:** 0.550 to 0.699
- **Low:** < 0.550
- **Priority:** High.

- **Verification:** Verified using boundary testing (e.g., inputting a score of 0.800 yields "Very High").

FR-4: Contextual Insights and Policy Recommendations

- **Description:** The system must display a list of actionable recommendations and policy strategies tailored to the country's predicted development tier.
- **Priority:** High.
- **Verification:** Inspected results page for tier-specific recommendations.

FR-5: Visual Progress Gauge

- **Description:** The results page must display a visual progress gauge (progress bar) that dynamically indicates the country's position on the human development spectrum.
- **Priority:** Medium.
- **Verification:** Checked that the gauge fill maps to the predicted score (e.g., a score of 0.75 fills the gauge by 75%).

FR-6: Input Range Validation

- **Description:** The input form must validate that user inputs fall within realistic ranges:
- **Life Expectancy:** 50 to 85 years
- **Expected Years of Schooling:** 5 to 20 years
- **Mean Years of Schooling:** 3 to 15 years
- **GNI per Capita:** \$500 to \$75,000
- **Priority:** Medium.
- **Verification:** Checked that the browser prevents form submission when inputs are out of bounds.

FR-7: Landing Page Dashboard

- **Description:** The application must provide a landing page explaining the three dimensions of HDI (Health, Education, and Income) to provide context for users.
- **Priority:** Medium.
- **Verification:** Page loads on the root '/' route.

3. Non-Functional Requirements (NFR)

NFR-1: Response Latency

- **Description:** The system must process inputs and render the prediction results page in under 2 seconds.
- **Priority:** High.
- **Verification:** Monitored network response times during prediction tests (<200ms).

NFR-2: Responsive Web Design

- **Description:** The interface must be responsive, rendering correctly on desktop, tablet, and mobile browsers.
- **Priority:** High.
- **Verification:** Tested layouts using Chrome DevTools responsive preview modes.

NFR-3: Model Accuracy

- **Description:** The machine learning model must achieve an $R^2 \geq 0.85$ on the test split.
- **Priority:** High.
- **Verification:** Checked training logs ($R^2 = 0.8834$).

NFR-4: Modern Visual Design

- **Description:** The interface must follow modern design principles, utilizing dark-mode glassmorphism cards and smooth transitions to improve user engagement.
- **Priority:** Medium.
- **Verification:** Visually inspected pages.

4. Software Environment Constraints

- **Python:** Version 3.10 or higher.
- **Framework:** Flask (WSGI framework).
- **Core Dependencies:** Scikit-learn, Pandas, NumPy, and Jinja2 templates.
- **Deployment:** Local web server on `http://127.0.0.1:5000/`.